

2. Let $\bar{M}(\pi) = \mathcal{N}(0, 1)$. Show that for all $\pi \in \Pi$, $v \in \Theta$, and for any choice of b ,

$$D_H^2(M_v(\pi), \bar{M}(\pi)) \leq \frac{1}{2} f_v^2(\pi) \leq \frac{(1-b)^2}{2} \mathbb{I}\{\langle v, \pi \rangle > b\},$$

i.e. information is obtained by the decision-maker only if the decision π aligns well with v in the model M_v .

3. Show that

$$\text{dec}_\gamma(\mathcal{M}, \bar{M}) \geq \inf_{p \in \Delta(\Pi)} \mathbb{E}_{v \sim \text{unif}(\mathbb{S}^{d-1})} \mathbb{E}_{\pi \sim p} \left[(1-b) - (1-b) \mathbb{I}\{\langle v, \pi \rangle > b\} - \gamma \frac{(1-b)^2}{2} \mathbb{I}\{\langle v, \pi \rangle > b\} \right].$$

4. Set $\varepsilon := 1 - b$. Use (6.75) and Part 3 above to argue that

$$\text{dec}_\gamma(\mathcal{M}', \bar{M}) \geq \varepsilon - \varepsilon \exp(-d/8) - \gamma \frac{\varepsilon^2}{2} \exp(-d/8).$$

Conclude that for $d \geq 8$,

$$\text{dec}_\gamma(\mathcal{M}', \bar{M}) \geq \frac{\varepsilon}{2} - \gamma \frac{\varepsilon^2}{2} \exp(-d/8)$$

5. Show that by choosing $\varepsilon = \frac{e^{d/8}}{6\gamma} \wedge \frac{1}{2}$ and recalling that $b = 1 - \varepsilon$, we get (6.74).

7. REINFORCEMENT LEARNING: FUNCTION APPROXIMATION AND LARGE STATE SPACES

In this section, we consider the problem of **online reinforcement learning with function approximation**. The framework is the same as that of Section 5 but, in developing algorithms, we no longer assume that the state and action spaces are finite/tabular, and in particular we will aim for "regret bounds" that are "independent of the number of states." To do this, we will make use of **function approximation**—either directly modeling the transition probabilities (for the underlying MDP), or modeling quantities such as value functions—and our goal will be to design algorithms that are capable of **generalizing across the state space** as they explore. This will pose challenges similar to that of the structured and contextual bandit settings, but we now face the additional challenge of credit assignment. Note that the online reinforcement learning framework is a special case of the general decision making setting in Section 6, but the algorithms we develop in this section will be tailored to the MDP structure.

Recall (Section 5) that for reinforcement learning, each MDP M takes the form

$$M = \{\underbrace{\mathcal{S}}_{\text{State Space}}, \underbrace{\mathcal{A}}_{\text{Action Space}}, \underbrace{\{P_h^M\}_{h=1}^H}_{\text{transition prob.}}, \underbrace{\{R_h^M\}_{h=1}^H}_{\text{reward dist.}}, \underbrace{d_1}_{\text{initial state dist.}}\},$$

where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $P_h^M : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$ is the probability transition kernel at step h , $R_h^M : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathbb{R})$ is the reward distribution, and $d_1 \in \Delta(\mathcal{S}_1)$ is the initial state distribution. All of the results in this section will take $\Pi = \Pi_{\text{RNS}}$, and we will assume that $\sum_{h=1}^H r_h \in [0, 1]$ unless otherwise specified.

7.1 Is Realizability Sufficient?

For the frameworks we have considered so far (contextual and structured bandits, general decision making), all of the algorithms we analyzed leveraged the assumption of **realizability**,

online RL with 함수 근사.
* 테이블 상 작은/유한한/tabular 크기의 state/action space를 가정하지 x.
* 함수 근사
① 기존의 MDP의 transition prob.를 직접적으로 계산하거나,
② Value function 같은 값을 직접적으로 계산.

"Realizability"

which asserts that we have a function class that is capable of modeling the underlying environment well. For reinforcement learning, there are various realizability assumptions one can consider:

- ① true MDP M^* ,
- ② optimal value function Q^{M^*} ,
- ③ optimal policy π_{M^*} 에 대한 "현실화 가능성" Realizability!

- *Model realizability*: We have a model class $\mathcal{M}$ of MDPs that contains the true MDP M^* .
- *Value function realizability*: We have a class $\mathcal{Q}$ of state-action value functions (Q -functions) that contains the optimal function $Q^{M^*,*}$ for the underlying MDP.
- *Policy realizability*: We have a class Π of policies that contains the optimal policy π_{M^*} .

Note that model realizability implies value function realizability, which in turn implies policy realizability. Ideally, we would like to be able to say that whenever one of these assumptions holds, we can obtain regret bounds that scale with the complexity of the function class (e.g., $\log|\mathcal{M}|$ for model realizability, or $\log|\mathcal{Q}|$ for value function realizability), but do not depend on the number of states $|S|$ or other properties of the underlying MDP, analogous to the situation for statistical learning. Unfortunately, the following result shows that this is too much to ask for.

Proposition 44 (e.g., Krishnamurthy et al. [55]): For any $S \in \mathbb{N}$ and $H \in \mathbb{N}$, there exists a class of horizon- H MDPs $\mathcal{M}$ with $|S| = S$, $|\mathcal{A}| = 2$, and $\log|\mathcal{M}| = \log(S)$, yet any algorithm must have

$$\mathbb{E}[\text{Reg}] \gtrsim \sqrt{\min\{S, 2^H\} \cdot T}.$$

state horizon

The interpretation of this result is that even if model realizability holds, any algorithm needs regret that scales with $\min\{|S|, |\mathcal{M}|, 2^H\}$. This means additional structural assumptions on the underlying MDP M^* —beyond realizability—are required if we want to obtain sample-efficient learning guarantees. (Note that since this construction satisfies model realizability, the strongest form of realizability, it also rules out sample-efficient results for value function and policy realizability.)

In what follows, we will explore different structural assumptions that facilitate low regret for reinforcement learning with function approximation. Briefly, the idea will be to make assumptions that either i) allow for extrapolation across the state space, or ii) control the number of "effective" state distributions the algorithm can encounter. We will begin by investigating "reinforcement learning" with linear models, then explore a general structural property known as Bellman rank.

Remark 19 (Comparison to structured bandits): Proposition 44 is analogous to the impossibility result we proved for structured bandits (Example 4.1), which is subsumed by the RL framework. That result required a large number of actions, while Proposition 44 holds even when $|\mathcal{A}| = 2$.

Remark 20 (Further notions of realizability): There are many notions of realizability beyond those we consider above. For example, for value function approximation, one can assume that $Q^{M^*, \pi} \in \mathcal{Q}$ for all π , (or assume that the class $\mathcal{Q}$ obeys certain notions of consistency with respect to the Bellman operator for M^*).

7.2 Linear Function Approximation

Toward understanding the complexity of RL with function approximation, let us consider perhaps the simplest possible modeling approach: Linear function approximation. A natural idea here is to assume linearity of the underlying Q -function corresponding to the true model M , generalizing the linear bandit setting in Section 4:

$$\text{Q-function linearity를 가정: } Q_h^{M, \star}(s, a) = \langle \phi(s, a), \theta_h^M \rangle, \quad \forall h \in [H] \quad (7.1)$$

$\phi(s, a)$: feature map known
 θ_h^M : unknown param. vector

where $\phi(s, a) \in \mathbb{B}_2^d(1)$ is a feature map that is known to the learner and $\theta_h^M \in \mathbb{B}_2^d(1)$ is an unknown parameter vector. Equivalently, we can define

$$\mathcal{Q} = \left\{ Q_h(s, a) = \langle \phi(s, a), \theta_h \rangle \mid \theta_h \in \mathbb{B}_2^d(1) \forall h \right\}, \quad (7.2)$$

class of state-action value functions
 어떤 선형적인 함수 그룹 전체

and assume that $Q^{M, \star} \in \mathcal{Q}$. This is called the Linear- Q^* model.

Linearity is a strong assumption, and it is reasonable to imagine that this would be sufficient for low regret. Indeed, one might hope that using linearity, we can extrapolate the value of $Q^{M, \star}$ once we estimate it for a small number of states. Unfortunately, even for this very simple class of functions, it turns out that realizability is still insufficient.

Proposition 45 (Weisz et al. [86], Wang et al. [85]): For any $d \in \mathbb{N}$ and $H \in \mathbb{N}$ sufficiently large, any algorithm for the Linear- Q^* model must have

$$\mathbb{E}[\text{Reg}] \gtrsim \min \left\{ 2^{\Omega(d)}, 2^{\Omega(H)} \right\}.$$

(This contrasts the situation for contextual bandits and linear bandits, where linear rewards were sufficient for low regret.) The intuition is that, even though $Q^{M, \star}$ is linear, it might take a very long time to estimate the value for even a small number of states. That is, linearity of the optimal value function is not a useful assumption unless there is some kind of additional structure that can guide us toward the optimal value function to being with.

We mention in passing that Proposition 45 can be proven by lower bounding the Decision-Estimation Coefficient [40].

The Low-Rank MDP model. Proposition 45 implies that linearity of the optimal Q -function alone is not sufficient for sample-efficient RL. To proceed, we will make a stronger assumption, which asserts that the transition probabilities themselves have linear structure. For all $s \in \mathcal{S}$, $a \in \mathcal{A}$, and $h \in [H]$, we have

$$P_h^M(s' | s, a) = \langle \phi(s, a), \mu_h^M(s') \rangle, \quad \text{and} \quad \mathbb{E}[r_h | s, a] = \langle \phi(s, a), w_h^M \rangle. \quad (7.3)$$

$\phi(s, a)$: feature map known
 $\mu_h^M(s')$: feature map unknown
 w_h^M : unknown param. vector.

Here, $\phi(s, a) \in \mathbb{B}_2^d(1)$ is a feature map that is known to the learner, $\mu_h^M(s') \in \mathbb{R}^d$ is another feature map which is unknown to the learner, and $w_h^M \in \mathbb{B}_2^d(\sqrt{d})$ is an unknown parameter

vector. (Additionally, for simplicity, we assume that $\|\sum_{s' \in \mathcal{S}} \mu_h^M(s')\| \leq \sqrt{d}$, which in particular holds in the tabular example below.) As before, assume that both cumulative and individual-step rewards are in $[0, 1]$. For the remainder of the subsection, we let $\mathcal{M}$ denote the set of MDPs with these properties.

$$\sum_{h=1}^H \mu_h \in [0, 1]$$

transition prob. linear.

→ The linear structure in (7.3) implies that the transition matrix has rank at most d , thus facilitating (as we shall see shortly) information sharing and generalization across states, (even when the cardinality of $\mathcal{S}$ and $\mathcal{A}$ is large or infinite.) For this reason, we refer to MDPs with this structure as low-rank MDPs [71, 88, 48, 5].

Just as linear bandits generalize unstructured multi-armed bandits, the low-rank MDP model (7.3) generalizes tabular RL, which corresponds to the special case in which $d = |\mathcal{S}| \cdot |\mathcal{A}|$, $\phi(s, a) = e_{s,a}$, and $(\mu_h(s'))_{s,a} = P_h^M(s' | s, a)$.

Properties of low-rank MDPs. The linear structure of the transition probabilities and mean rewards is a significantly more stringent assumption than linearity of $Q_h^{M,*}(s, a)$ in (7.1). Notably, it implies that Bellman backups of arbitrary functions are linear.

Linearity of Bellman operator.

• 각 상태의 가치를 구하는 Bell operator의 선형성.

• 개기전 특장: 가치 함수 $V(\pi)$ 는 해당 상태에서부터 ~ 앞으 모든 모든 리워드의 합

$$V(\pi) = R_0 + \gamma \sum_{s \in \mathcal{S}} P(s | s_0) V(s)$$

→ 즉, 이전에는 모든 상태와 가능한 모든 전이"를 고려한다.

→ 1) 직접 계산하기 어렵.

→ 2) 굳이?

점진적 통해 가치 함수를 점진적으로 개선해 나가는데, 굳이 모든 가능한 전이를 고려할 필요를 왜 없을까!

Lemma 25: For any low-rank MDP $M \in \mathcal{M}$ and any $Q : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ and any $h \in [H]$, the Bellman operator is linear in ϕ :

Bellman operator는 (이제) 선형.

Bellman Operator.

$$[\mathcal{T}_h^M Q](s, a) = \langle \phi(s, a), \theta_Q^M \rangle$$

벨만 op.의 선형성!

for some $\theta_Q^M \in \mathbb{R}^d$. (In particular, this implies that for any policy $\pi = (\pi_1, \dots, \pi_H)$, functions $Q_h^{M,\pi}$ are linear (in ϕ) for every h .) Finally, for $Q : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$, it holds that

$$\|\theta_Q^M\| \leq 2\sqrt{d}. \quad \theta_Q^M = [\mathcal{T}_h^M Q_h^{M,\pi}] \text{ 대역에 대응}$$

즉, (low-rank MDP)에 대해, 모든 정책 π 에 대해 $Q_h^{M,\pi}$ 는 선형이다!

As a special case, this lemma implies that for low-rank MDPs, $Q_h^{M,\pi}$ is linear for all π .

Proof of Lemma 25. We have

$$[\mathcal{T}_h^M Q](s, a) = \langle \phi(s, a), w_h^M \rangle + \sum_{s'} P_h^M(s' | s, a) \max_{a'} Q(s', a') \quad (7.4)$$

$$= \langle \phi(s, a), w_h^M \rangle + \sum_{s'} \langle \phi(s, a), \mu_h^M(s') \rangle \max_{a'} Q(s', a') \quad (7.5)$$

$$= \left\langle \phi(s, a), \left(w_h^M + \sum_{s'} \mu_h^M(s') \max_{a'} Q(s', a') \right) \right\rangle = \langle \phi(s, a), \theta_Q^M \rangle \quad (7.6)$$

✓ The second statement follows since $Q_h^{M,\pi} = [\mathcal{T}_h^M Q_{h+1}^{M,\pi}]$. For the last statement, $\|\theta_Q^M\| \leq 2\sqrt{d}$

$$\|\theta_Q^M\| \leq \|w_h^M\| + \left\| \sum_{s'} \mu_h^M(s') Q(s') \right\| \leq 2\sqrt{d}, \quad \begin{aligned} & \text{(def.)} \\ & \mu_h^M(s') \in \mathbb{R}^d \quad \|\sum_{s \in \mathcal{S}} \mu_h^M(s)\| \leq \sqrt{d} \\ & w_h^M \in \mathbb{B}_2^d(\sqrt{d}) \end{aligned} \quad (7.7)$$

since μ_h^M is a vector of distributions on $\mathcal{S}$. □

7.2.1 The LSVI-UCB Algorithm

To provide regret bounds for the low-rank MDP model, we analyze an algorithm called LSVI-UCB ("Least Squares Value Iteration UCB"), which was introduced and analyzed in

Least Squares Value Iteration - UCB

the influential paper of Jin et al. [48]. Similar to the UCB-VI algorithm we analyzed for tabular RL, the main idea behind the algorithm is to compute a **state-action value** $\bar{Q}^t$ with the "optimistic property" that

$$\bar{Q}_h^t(s, a) \geq Q_h^{M, \star}(s, a) \quad \text{66} \quad \text{79} \quad \text{라는 optimistic property를 가지고 있는, state-action value 함수를 계산!}$$

for all s, a, h . This is achieved by combining the principle of **dynamic programming** with an appropriate choice of **bonus** (to ensure optimism.) However, unlike UCB-VI, the algorithm does **not directly estimate transition probabilities** (which is not feasible when μ^M is unknown), and instead implements **approximate value iteration** by solving a certain least squares objective.

transition prob. (p)를
직접적으로 계산하지 않음.
대신, approximate VI

LSVI-UCB *Least Square Value Iteration*

Input: $R, \rho > 0$

for $t = 1, \dots, T$ **do**

Let $\bar{Q}_{H+1}^t \equiv 0$. *State-action value*

for $h = H, \dots, 1$ **do**

Compute least-squares estimator

$$\hat{\theta}_h^t = \arg \min_{\theta \in \mathcal{B}_2^d(\rho)} \sum_{i < t} \left(\langle \phi(s_h^i, a_h^i), \theta \rangle - r_h^i - \max_a \bar{Q}_{h+1}^t(s_{h+1}^i, a) \right)^2,$$

and let $\hat{Q}_h^t(s, a) := \langle \phi(s, a), \hat{\theta}_h^t \rangle$.
Define

$$\Sigma_h^t = \sum_{i < t} \phi(s_h^i, a_h^i) \phi(s_h^i, a_h^i)^\top + I. \quad \text{elliptic bonus}$$

Compute bonus:

$$b_{h,\delta}^t(s, a) = \sqrt{R} \|\phi(s, a)\|_{(\Sigma_h^t)^{-1}}. \quad \text{to ensure optimism}$$

Compute optimistic value function:

$$\bar{Q}_h^t(s, a) = \left\{ \hat{Q}_h^t(s, a) + b_{h,\delta}^t(s, a) \right\} \wedge 1. \quad \text{ensures optimism of } \bar{Q}^t \text{ (state-action value function)}$$

Set $\bar{V}_h^t(s) = \max_{a \in \mathcal{A}} \bar{Q}_h^t(s, a)$ and $\hat{\pi}_h^t(s) = \arg \max_{a \in \mathcal{A}} \bar{Q}_h^t(s, a)$.

Collect trajectory $(s_1^t, a_1^t, r_1^t), \dots, (s_H^t, a_H^t, r_H^t)$ according to $\hat{\pi}^t$.

In more detail, for each episode t , the algorithm computes $\bar{Q}_1^t, \dots, \bar{Q}_H^t$ through approximate dynamic programming. At layer h , given $\bar{Q}_{h+1}^t$, the algorithm computes a **linear Q-function** $\hat{Q}_h^t(s, a) := \langle \phi(s, a), \hat{\theta}_h^t \rangle$, by solving a least squares problem in which $X = \phi(s_h, a_h)$ is the **feature vector** and $Y = r_h + \max_a \bar{Q}_{h+1}^t(s_{h+1}, a)$ is the **target/outcome**. (This is motivated by **Lemma 25**, which asserts that the Bellman backup $[\mathcal{T}_h^M \bar{Q}_{h+1}^t](s, a)$ is linear.) Given $\hat{Q}_h^t$, the algorithm forms the optimistic estimate $\bar{Q}_h^t$ via

$$\bar{Q}_h^t(s, a) = \left\{ \hat{Q}_h^t(s, a) + b_{h,\delta}^t(s, a) \right\} \wedge 1,$$

where

bonus

$$b_{h,\delta}^t(s, a) = \sqrt{R} \|\phi(s, a)\|_{(\Sigma_h^t)^{-1}}, \quad \text{with } \Sigma_h^t = \sum_{i < t} \phi(s_h^i, a_h^i) \phi(s_h^i, a_h^i)^\top + I,$$

(defined)
elliptic bonus

Lemma 15 (Error decomposition for optimistic policies): Let $\{\bar{Q}_1, \dots, \bar{Q}_H\}$ be a sequence of functions $\bar{Q}_h: \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ with the property that for all (s, a) ,
 (Sequence of optimistic value functions)
 $\bar{Q}_h^{M,*}(s, a) \leq \bar{Q}_h(s, a)$ (5.22)
 and set $\bar{Q}_{H+1} \equiv 0$. Let $\hat{\pi} = (\hat{\pi}_1, \dots, \hat{\pi}_H)$ be such that $\hat{\pi}_h(s) = \arg \max_a \bar{Q}_h(s, a)$. Then for all $s \in \mathcal{S}$,
 $V_1^{M,*}(s) - V_1^{M,\hat{\pi}}(s) \leq \sum_{h=1}^H \mathbb{E}^{M,\hat{\pi}}[(\bar{Q}_h - \mathcal{T}_h^M \bar{Q}_{h+1})(s_h, \hat{\pi}(s_h)) \mid s_1 = s]$. (5.23)
 Lemma 15 tells us that closeness of $\bar{Q}_h$ to the Bellman backup $\mathcal{T}_h^M \bar{Q}_{h+1}$ implies closeness of $\hat{\pi}$ to π_* (in terms of the value)

is an elliptic bonus (analogous to the bonus used within LinUCB.) With this, the algorithm proceeds to the next layer $h-1$. Once $\bar{Q}^t$ is computed for every layer, the algorithm executes the optimistic policy $\hat{\pi}^t$ given by $\hat{\pi}_h^t(s) = \arg \max_{a \in \mathcal{A}} \bar{Q}_h^t(s, a)$.

The LSVI-UCB algorithm enjoys the following regret bound.

Proposition 46: If any $\delta > 0$, if we set $R = c \cdot d^2 \log(HT/\delta)$ (for a sufficiently large numerical constant c) and $\rho = 2\sqrt{d}$, LSVI-UCB has that with probability at least $1 - \delta$,

$$\text{Reg} \lesssim H \sqrt{d^3 \cdot T \log(HT/\delta)}. \quad (7.8)$$

7.2.2 Proof of Proposition 46

The starting point of our analysis for UCB-VI was Lemma 15, which states that it is sufficient to construct optimistic estimates $\{\bar{Q}_1, \dots, \bar{Q}_H\}$ (i.e. $\bar{Q}_h^{M,*} \leq \bar{Q}_h$) such that the Bellman residuals $\mathbb{E}^{M,\hat{\pi}}[(\bar{Q}_h - \mathcal{T}_h^M \bar{Q}_{h+1})(s_h, a_h)]$ are small under the greedy (with respect to $\bar{Q}$'s) policy $\hat{\pi}$. In order to control these residuals, we constructed an estimated model $\hat{M}$ and defined empirical Bellman operators $\mathcal{T}_h^M$ in terms of estimated transition kernels. We then set $\bar{Q}_h$ to be the empirical Bellman backup $\mathcal{T}_h^M \bar{Q}_{h+1}$, plus an optimistic bonus term. In contrast, LSVI-UCB does not directly estimate the model. Instead, it performs regression with a target that is an empirical Bellman backup. As we shall see shortly, subtleties arise in the analysis of this regression step due to lack of independence.

Technical lemmas for regression. Recall from Lemma 25 that for any fixed $Q: \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$,
 Bellman Operator의 선형성.

$$\mathbb{E}^M \left[r_h^i + \max_a Q(s_{h+1}^i, a) \mid s_h^i, a_h^i \right] = [\mathcal{T}_h^M Q](s_h^i, a_h^i). \quad (7.9)$$

However, for layer h , the regression problem within LSVI-UCB concerns a data-dependent function $Q = \bar{Q}_{h+1}^t$ (with $i < t$), which is chosen as a function of all the trajectories $\tau^1, \dots, \tau^{t-1}$. This dependence implies that the regression problem solved by LSVI-UCB is not of the type studied, say, in Proposition 1. Instead, in the language of Section 1.4, the mean of the outcome variable is itself a function that depends on all the data. The saving grace here is that this dependence does not result in arbitrarily complex functions, which will allow us to appeal to uniform convergence arguments. In particular, for every h and t , $\bar{Q}_h^t$ belongs to the class

$$\bar{Q}_h^t \text{ belongs to } \dots \quad \mathcal{Q} := \left\{ (s, a) \mapsto \left\{ \langle \theta, \phi(s, a) \rangle + \sqrt{R} \|\phi(s, a)\|_{(\Sigma)^{-1}} \right\} \wedge 1 : \|\theta\| \leq 2\sqrt{d}, \sigma_{\min}(\Sigma) \geq 1 \right\}. \quad (7.10)$$

To make use of this fact, we first state an abstract result concerning regression with dependent outcomes.

Lemma 26: Let $\mathcal{G}$ be an abstract set with $|\mathcal{G}| < \infty$. Let $x_1, \dots, x_T \in \mathcal{X}$ be fixed, and for each $g \in \mathcal{G}$, let $y_1(g), \dots, y_T(g) \in \mathbb{R}$ be 1-subGaussian outcomes satisfying

$$\mathbb{E}[y_i(g) \mid x_i] = f_g(x_i)$$

for $f_g \in \mathcal{F} \subseteq \{f : \mathcal{X} \rightarrow \mathbb{R}\}$.^a In addition, assume that $y_1(g), \dots, y_T(g)$ are conditionally independent given $x_1, \dots, x_T$. For any latent $g \in \mathcal{G}$, define the least-squares solution

$$\hat{f}_g \in \arg \min_{f \in \mathcal{F}} \sum_{i=1}^T (y_i(g) - f(x_i))^2.$$

즉,

With probability at least $1 - \delta$, simultaneously for all $g \in \mathcal{G}$,

Lemma 26. 즉.

$$\sum_{i=1}^T (\hat{f}_g(x_i) - f_g(x_i))^2 \lesssim \log(|\mathcal{F}||\mathcal{G}|/\delta).$$

^aThe random variables $\{y_i(g)\}_{g \in \mathcal{G}}$ may be correlated.

Proof of Lemma 26. Fix $g \in \mathcal{G}$. To shorten the notation, it is useful to introduce empirical norms $\|f\|_T^2 = \frac{1}{T} \sum_{i=1}^T f(x_i)^2$ and empirical inner product $\langle f, f' \rangle_T = \sum_{i=1}^T f(x_i)f'(x_i)$ for $f, f' \in \mathcal{F}$. Optimality of $\hat{f}_g$ implies that

$$\sum_{i=1}^T (y_i(g) - \hat{f}_g(x_i))^2 \leq \sum_{i=1}^T (y_i(g) - f_g(x_i))^2$$

which can be written succinctly (with a slight abuse of notation) as $\|Y_g - \hat{f}_g\|_T^2 \leq \|Y_g - f_g\|_T^2$ for $Y_g = (y_1(g), \dots, y_T(g))$. This implies

$$\|\hat{f}_g - f_g\|_T^2 \leq 2 \langle Y_g - f_g, \hat{f}_g - f_g \rangle_T.$$

Dividing both sides by $\|\hat{f}_g - f_g\|_T$ and taking supremum over $\hat{f}_g \in \mathcal{F}$ leads to

$$\|\hat{f}_g - f_g\|_T \leq 2 \max_{f \in \mathcal{F}} \langle Y_g - f_g, \frac{f - f_g}{\|f - f_g\|_T} \rangle_T. \quad (7.11)$$

The random vector $Y_g - f_g$ has independent zero-mean 1-subGaussian entries by assumption, while the multiplier $\frac{f - f_g}{\|f - f_g\|_T}$ is simply a T -dimensional vector of Euclidean length $\sqrt{T}$, for each $f \in \mathcal{F}$. Hence, each inner product in (7.11) is a **sub-Gaussian vector** with variance proxy $\frac{1}{T}$ (see Definition 2). Thus, with probability at least $1 - \delta$, the maximum on the right-hand side does not exceed $C \sqrt{\log(|\mathcal{F}|/\delta)/T}$ for an appropriate constant C . Taking the union bound over g and squaring both sides of (7.11) yields the desired bound. $\square$

We may now apply Lemma 26 to analyze the regression step of LSVI-UCB.

Lemma 27: (With probability at least $1 - \delta$) we have that for all t and h ,

$$\sum_{i < t} \left(\hat{Q}_h^t(s_h^i, a_h^i) - [\mathcal{T}_h^M \bar{Q}_{h+1}^t](s_h^i, a_h^i) \right)^2 \lesssim d^2 \log(HT/\delta). \quad (7.12)$$

Proof sketch for Lemma 27. Let $t \in [T]$ and $h \in [H]$ be fixed. To make the correspondence with Lemma 26 explicit, for the data $(s_h^i, a_h^i, s_{h+1}^i, r_h^i)$, we define $x_i = \phi(s_h^i, a_h^i)$ and $y_i(Q) = r_h^i + \max_a Q(s_{h+1}^i, a)$, with $Q \in \mathcal{Q}$ playing the role of the index $g \in \mathcal{G}$. With this, we have

$$\mathbb{E}[y_i(Q) \mid x_i] = \mathbb{E}^M \left[r_h^i + \max_a Q(s_{h+1}^i, a) \mid s_h^i, a_h^i \right] = [\mathcal{T}_h^M Q](s_h^i, a_h^i) = \langle \phi(s_h^i, a_h^i), \theta_Q^M \rangle$$

which is linear in $x^i = \phi(s_h^i, a_h^i)$, with the vector of coefficients θ_Q^M depending on Q . The regression problem is well-specified as long as we choose

$$\mathcal{F} = \left\{ \phi(s, a) \mapsto \langle \phi(s, a), \theta \rangle : \|\theta\| \leq 2\sqrt{d} \right\}$$

and Q as in (7.10). While both of these sets are infinite, we can to a standard covering number argument for an appropriate scale ε . The cardinalities of ε -discretized classes can be shown to be of size $\tilde{O}(d)$ and $\tilde{O}(d^2)$, respectively, up to factors logarithmic in $1/\varepsilon$ and d . The statement follows after checking that discretization incurs a small price due to Lipschitzness with respect to parameters. Finally, we union bound over t and h . $\square$

Establishing optimism. The next lemma shows that closeness of the regression estimate to the Bellman backup on the data $\{(s_h^i, a_h^i)\}_{i < t}$ translates into closeness at an arbitrary (s, a) pair as long as $\phi(s, a)$ is sufficiently covered by the data collected so far. This, in turn, implies that $\bar{Q}_1^t, \dots, \bar{Q}_H^t$ are optimistic.

Lemma 28: Whenever the event in Lemma 27 occurs, we have that for all (s, a, h) and $t \in [T]$,

$$\textcircled{1} \quad \left| \bar{Q}_h^t(s, a) - [\mathcal{T}_h^M \bar{Q}_{h+1}^t](s, a) \right| \leq \sqrt{d^2 \log(HT/\delta)} \cdot \|\phi(s, a)\|_{(\Sigma_h^t)^{-1}} =: b_{h,\delta}^t(s, a). \quad (7.13)$$

and LHO: Lemma 27의 식과 내부의 각 (사실상의) term

$$\textcircled{2} \quad \bar{Q}_h^t(s, a) \geq Q_h^{M,*}(s, a). \quad (7.14)$$

optimistic property

Proof of Lemma 28. Writing the Bellman backup, via Lemma 25, as

$$[\mathcal{T}_h^M \bar{Q}_{h+1}^t](s, a) = \langle \phi(s, a), \theta_h^t \rangle \quad \hookrightarrow [\mathcal{T}_h^M Q](s, a) = \langle \phi(s, a), \theta_h^M \rangle$$

Bellman operator which is linear in ϕ .

(for some $\theta_h^t \in \mathbb{R}^d$ with $\|\theta_h^t\|_2 \leq 2\sqrt{d}$) we have that

$$\begin{aligned} \left| \bar{Q}_h^t(s, a) - [\mathcal{T}_h^M \bar{Q}_{h+1}^t](s, a) \right| &= \left| \langle \phi(s, a), \hat{\theta}_h^t - \theta_h^t \rangle \right| \\ &= \left| \langle (\Sigma_h^t)^{-1/2} \phi(s, a), (\Sigma_h^t)^{1/2} (\hat{\theta}_h^t - \theta_h^t) \rangle \right| \\ &\leq \sqrt{d^2 \log(HT/\delta)} \cdot \|\hat{\theta}_h^t - \theta_h^t\|_{\Sigma_h^t} \end{aligned}$$

LHO: Lemma 27의 식과 내부의 각 (사실상의) term

Lemma 27 then implies (7.13), since

$$\|\hat{\theta}_h^t - \theta_h^t\|_{\Sigma_h^t}^2 = (\hat{\theta}_h^t - \theta_h^t)^\top \left(\sum_{i < t} \phi(s_h^i, a_h^i) \phi(s_h^i, a_h^i)^\top + I \right) (\hat{\theta}_h^t - \theta_h^t) \quad (7.15)$$

decompose

$$= \sum_{i < t} \left(\bar{Q}_h^t(s_h^i, a_h^i) - [\mathcal{T}_h^M \bar{Q}_{h+1}^t](s_h^i, a_h^i) \right)^2 + \|\hat{\theta}_h^t - \theta_h^t\|^2 \quad (7.16)$$

and $\|\hat{\theta}_h^t - \theta_h^t\|^2 \leq d$ by (7.7).

To show (7.14), we proceed by induction on $\bar{V}_h^t \geq V_h^{M,*}$, as in the proof of Lemma 16. We start with the base case $h = H+1$, which has $\bar{V}_{H+1}^t = V_{H+1}^{M,*} \equiv 0$. Assuming

Some MDP families...

- Low-Rank MDPs [87, 48, 5]
- MDPs with low occupancy complexity [34].
- Block MDPs and reactive POMDPs [55, 33].
- Linear mixture MDPs [65, 15].
- MDPs with Linear Q^* and V^* [34].
- Linear dynamical systems (LQR) [30].

We will learn about these examples in [Section 7.3.3](#).



Building intuition. Bellman rank is a property (of the underlying MDP M^*) which gives a way of controlling *distribution shift*—that is, how many times a deliberate algorithm can be surprised by a substantially new state distribution $d^{M,\pi}$ (when it updates its policy). To motivate the property, let us revisit the low-rank MDP model. Let M be a low-rank MDP with feature map $\phi(s, a) \in \mathbb{R}^d$, and let $Q_h(s, a) = \langle \phi(s, a), \theta_h^Q \rangle$ be an arbitrary linear value function. Observe that since M is a Low-Rank MDP, we have $[\mathcal{T}_h^M Q](s, a) = \langle \phi(s, a), \tilde{\theta}_h^{M,Q} \rangle$, where $\tilde{\theta}_h^{M,Q} := w_h^M + \int \mu_h^M(s') \max_{a'} Q_{h+1}(s', a') ds'$. As a result, for any policy π , we can write the Bellman residual for Q as

$$\mathbb{E}^{M,\pi} [Q_h(s_h, a_h) - r_h - \max_a Q_{h+1}(s_{h+1}, a)] = \langle \mathbb{E}^{M,\pi} [\phi(s_h, a_h)], \theta_h^Q - w_h^M - \tilde{\theta}_h^{M,Q} \rangle \quad (7.20)$$

where $X_h^M(\pi) := \mathbb{E}^{M,\pi} [\phi(s_h, a_h)] \in \mathbb{R}^d$ is an “embedding” that depends on π but not Q , and $W_h^M(Q) := \theta_h^Q - w_h^M - \tilde{\theta}_h^{M,Q} \in \mathbb{R}^d$ is an embedding that depends on Q but not π (both embeddings depend on M). Notably, if we view the Bellman residual as a huge $\Pi \times \mathcal{Q}$ matrix $\mathcal{E}_h(\cdot, \cdot) \in \mathbb{R}^{\Pi \times \mathcal{Q}}$ with

$$\mathcal{E}_h(\pi, Q) := \mathbb{E}^{M,\pi} [Q_h(s_h, a_h) - (r_h + \max_a Q_{h+1}(s_{h+1}, a))], \quad (7.22)$$

then the property (7.21) implies that $\text{rank}(\mathcal{E}_h(\cdot, \cdot)) \leq d$. Bellman rank is an abstraction of this property.¹⁶

Definition 8 (Bellman rank): For an MDP M with value function class $\mathcal{Q}$ and policy class Π , the Bellman rank is defined as

$$d_B(M) = \max_{h \in [H]} \text{rank}(\{\mathcal{E}_h(\pi, Q)\}_{\pi \in \Pi, Q \in \mathcal{Q}}). \quad (7.23)$$

Equivalently, Bellman rank is the smallest dimension d such that for all h , there exist embeddings $X_h^M(\pi), W_h^M(Q) \in \mathbb{R}^d$ such that

$$\mathcal{E}_h(\pi, Q) = \langle X_h^M(\pi), W_h^M(Q) \rangle \quad (7.24)$$

for all $\pi \in \Pi$ and $Q \in \mathcal{Q}$.

¹⁶Bellman rank was originally introduced in the pioneering work of Jiang et al. [46]. The definition of Bellman rank we present, which is slightly different from the original definition, is taken from the later work of Du et al. [34], Jin et al. [49], and is often referred to as Q -type Bellman rank.

The utility of Bellman rank is that the factorization in (7.24) gives a way of controlling distribution shift in the MDP M , which facilitates the application of standard generalization guarantees for supervised learning/estimation. Informally, there are only d effective directions (in which we can be “surprised” by the state distribution induced by a policy π), to the extent that this matters for the class $\mathcal{Q}$ under consideration; this property was used implicitly in the proof of the regret bound for LSVI-UCB. (As we will see, low Bellman rank is satisfied in many settings that go beyond the Low-Rank MDP model.)

7.3.1 The BiLinUCB Algorithm

We now present an algorithm, BiLinUCB [34], which attains low regret for MDPs with low Bellman rank (under the realizability assumption that

$$Q^{M^*,*} \in \mathcal{Q}.$$

Like many of the algorithms we have covered, BiLinUCB is based on confidence sets and optimism, though the way we will construct the confidence sets and implement optimism is new.

PAC versus regret. For technical reasons, we will not directly give a regret bound for BiLinUCB. Instead, we will prove a PAC (“probably approximately correct”) guarantee. For PAC, the algorithm plays for T episodes, then outputs a final policy $\hat{\pi}$, and its performance is measured via

$$\|f^{M^*}(\pi_{M^*}) - f^{M^*}(\hat{\pi}).\| \quad (7.25)$$

That is, instead of considering cumulative performance (as with regret), we are only concerned with final performance. For PAC, we want to ensure that $f^{M^*}(\pi_{M^*}) - f^{M^*}(\hat{\pi}) \leq \varepsilon$ for some $\varepsilon \ll 1$ using a number of episodes that is polynomial in ε^{-1} and other problem parameters. This is an easier task (than achieving low regret). If we have an algorithm that ensures that $\mathbb{E}[\mathbf{Reg}] \lesssim \sqrt{CT}$ for some problem-dependent constant C , we can turn this into an algorithm that achieves PAC error ε using $O(\frac{C}{\varepsilon^2})$ episodes via online-to-batch conversion. In the other direction, if we have an algorithm that achieves PAC error ε using $O(\frac{C}{\varepsilon^2})$ episodes, one can use this to achieve $\mathbb{E}[\mathbf{Reg}] \lesssim C^{1/3}T^{2/3}$ using a simple explore-then-commit approach; this is lossy, but is the best one can hope for in general.

Algorithm overview. BiLinUCB proceeds in K iterations, each of which consists of n episodes. The algorithm maintains a confidence set $\mathcal{Q}^k \subseteq \mathcal{Q}$ of value functions (generalizing the confidence sets we constructed for structured bandits in Section 4), with the property that $Q^{M^*,*} \in \mathcal{Q}$ with high probability. Each iteration k consists of two parts:

- Given the current policy confidence set $\mathcal{Q}^k$, the algorithm computes a value function Q^k and corresponding policy $\pi^k := \pi_{Q^k}$ that is optimistic on average

각 (K번) iteration마다 주어진 confidence set ($\mathcal{Q}^k$ 에 대해, * Value func. Q^k 를 계산하고 * 그에 상응하는 정책 $\pi^k (= \pi_{Q^k})$ 를 계산. 이때, 그 정책은 optimistic함.

$$Q^k = \arg \max_{Q \in \mathcal{Q}^k} \mathbb{E}_{s_1 \sim d_1} [Q_1(s_1, \pi_Q(s_1))].$$

The main novelty here is that we are only aiming for optimism with respect to the initial state distribution.

→ 만능인 M3은 policy π^k 를 가지고
 n 개의 episode 생성,

Bellman residual을 리스하는
 estimator인
 $\{\hat{\mathcal{E}}_h^k(Q)\}_{h \in [H]}$ 을 계산.
 ↓
 정책 $\pi^1, \dots, \pi^k$ 에서
 그 estimator의 값이 작도록 하는
 Value function을 남으로 제형해서,
 ⇒ M3은 (confidence set $\mathcal{Q}$ 을 만들).

즉, Bellman Residual 값이 큰
 Value function은 제거함!
 (우리가 목표 하는 $Q^{M^*,*}$ 이
 zero Bellman Residual값을 가짐.)

• Using the new policy π^k the algorithm gathers n episodes and uses these to compute estimators $\{\hat{\mathcal{E}}_h^k(Q)\}_{h \in [H]}$ (which approximate the Bellman residual $\mathcal{E}_h(\pi^k, Q)$ for all $Q \in \mathcal{Q}$). Then, in (7.26), the algorithm computes the new confidence set $\mathcal{Q}^{k+1}$ by restricting to value functions (for which the estimated Bellman residual is small for $\pi^1, \dots, \pi^k$). Eliminating value functions with large Bellman residual is a natural idea, because we know from the Bellman equation that $Q^{M^*,*}$ has zero Bellman residual.

BiLinUCB
 Input: $\beta > 0$, iteration count $K \in \mathbb{N}$, batch size $n \in \mathbb{N}$.
 $\mathcal{Q}^1 \leftarrow \mathcal{Q}$.
for iteration $k = 1, \dots, K$ **do** (Iterations)
 Compute optimistic value function:
 (optimistic on average)
 $Q^k = \arg \max_{Q \in \mathcal{Q}^k} \mathbb{E}_{s_1 \sim d_1} [Q_1(s_1, \pi_Q(s_1))]$.

 and let $\pi^k := \pi_{Q^k}$. (이 Value function을 바탕으로 new policy)
 for $l = 1, \dots, n$ **do** (n episodes, for each Iteration)
 Execute π^k (for an episode) and observe trajectory $(s_1^{k,l}, a_1^{k,l}, r_1^{k,l}), \dots, (s_H^{k,l}, a_H^{k,l}, r_H^{k,l})$.
 Compute confidence set
 new

$$\mathcal{Q}^{k+1} = \left\{ Q \in \mathcal{Q} \mid \sum_{i \leq k} (\hat{\mathcal{E}}_h^i(Q))^2 \leq \beta \quad \forall h \in [H] \right\}, \quad (7.26)$$

 where
 estimator for Bellman residual.
 Bellman Residual에 대한 estimator가

$$\hat{\mathcal{E}}_h^i(Q) := \frac{1}{n} \sum_{l=1}^n \left(Q_h(s_h^{i,l}, a_h^{i,l}) - r_h^{i,l} - \max_{a \in \mathcal{A}} Q_{h+1}(s_{h+1}^{i,l}, a) \right).$$

 Let $\hat{k} = \arg \max_{k \in [K]} \hat{V}^k$, where $\hat{V}^k := \frac{1}{n} \sum_{l=1}^n \sum_{h=1}^H r_h^{k,l}$.
 Return $\hat{\pi} = \pi^{\hat{k}}$.

Main guarantee. The main result for this section is the following **PAC** guarantee for BiLinUCB. (probably Approximately correct (작은 Reg와 비슷한 의미))



Proposition 47: Suppose that M^* has Bellman rank d and $Q^{M^*,*} \in \mathcal{Q}$. For any $\varepsilon > 0$ and $\delta > 0$, (if we set $n \gtrsim \frac{H^3 d \log(|\mathcal{Q}|/\delta)}{\varepsilon^2}$, $K \gtrsim H d \log(1+n/d)$, and $\beta \propto c \cdot K \frac{\log |\mathcal{Q}| + \log(HK/\delta)}{n}$), then BiLinUCB learns a policy $\hat{\pi}$ such

algorithm

policy learned by BiLinUCB

$f^{M^*}(\pi_{M^*}) - f^{M^*}(\hat{\pi}) \leq \varepsilon$ **PAC guarantee!!** of the algorithm BiLinUCB

with probability at least $1 - \delta$, and does so using

$$\tilde{O}\left(\frac{H^4 d^2 \log(|\mathcal{Q}|/\delta)}{\varepsilon^2}\right)$$

이 정도는 4.

episodes.)

This result shows that low Bellman rank suffices to learn a near-optimal policy, with sample complexity that only depends on the rank d , the horizon H , and the capacity $\log|Q|$ for the value function class; this reflects that the algorithm is able to generalize across the state space, with d and $\log|Q|$ controlling the degree of generalization. The basic principles at play are: rank d & $\log|Q|$ "degree of generalization control!"

- By choosing Q^k ^(optimistic or average) optimistically, we ensure that the suboptimality of the algorithm is controlled by the Bellman residual for Q^k , *on-policy*, similar to what we saw for UCB-VI and LSVI-UCB. An important difference compared to the LSVI-UCB algorithm we covered in the previous section is that BiLinUCB is only optimistic "on average" with respect to the initial state distribution, i.e.,

$$\mathbb{E}_{s_1 \sim d_1} [Q_1^k(s_1, \pi_{Q^k}(s_1))] \geq f^{M^*}(\pi_{M^*}),$$

(while LSVI-UCB aims to find a value function that is uniformly optimistic for all states and actions.)

- The confidence set construction (7.26) explicitly removes value functions that have large Bellman residual on the policies encountered so far. The key role of the Bellman rank property is to ensure that there are only $\tilde{O}(d)$ "effective" state distributions that lead to substantially different values for the Bellman residual, which means that eventually, only value functions (with low residual) will remain.

Interestingly, the Bellman rank property is only used for analysis, and the algorithm does not explicitly compute or estimate the factorization.

Regret bounds. The BiLinUCB algorithm can be lifted to provide a regret guarantee via a explore-then-commit strategy: Run the algorithm for T_0 episodes to learn a ε -optimal policy, then commit to this policy for the remaining rounds. It is a simple exercise to show that by choosing T_0 appropriately, this approach gives

$$\text{Reg} \leq \tilde{O}\left((H^4 d^2 \log(|Q|/\delta))^{1/3} \cdot T^{2/3}\right).$$

(Under an additional assumption known as *Bellman completeness*, it is possible to attain $\sqrt{T}$ with a variant of this algorithm that uses a slightly different confidence set construction [49].)

7.3.2 Proof of Proposition 47

Recall from the definition of Bellman rank that there exist embeddings $X_h^{M^*}(\pi), W_h^{M^*}(Q) \in \mathbb{R}^d$ such that for all $\pi \in \Pi$ and $Q \in \mathcal{Q}$,

$$\mathcal{E}_h(\pi, Q) = \langle X_h^{M^*}(\pi), W_h^{M^*}(Q) \rangle.$$

We assume throughout this proof that $\|X_h^{M^*}(\pi)\|, \|W_h^{M^*}(Q)\|_2 \leq 1$ for simplicity.

Technical lemmas. Before proceeding, we state two technical lemmas. The first lemma establishes validity for the confidence set $\mathcal{Q}^k$ constructed by BiLinUCB.

Lemma 29: For any $\delta > 0$, if we set $\beta = c \cdot K \frac{\log|\mathcal{Q}| + \log(HK/\delta)}{n}$, where $c > 0$ is sufficiently large absolute constant, then with probability at least $1 - \delta$, for all $k \in [K]$:

1. All $Q \in \mathcal{Q}^k$ have

$$\sum_{i < k} (\mathcal{E}_h(\pi^i, Q))^2 \lesssim \beta \quad \forall h \in [H]. \quad (7.27)$$

2. $Q^{M^*, \star} \in \mathcal{Q}^k$.

Proof of Lemma 29. Using Hoeffding's inequality and a union bound (Lemma 3), we have that with probability at least $1 - \delta$, for all $k \in [K]$, $h \in [H]$, and $Q \in \mathcal{Q}$,

$$\left| \widehat{\mathcal{E}}_h^k(Q) - \mathcal{E}_h(\pi^k, Q) \right| \leq C \cdot \sqrt{\frac{\log(|\mathcal{Q}|HK/\delta)}{n}}, \quad (7.28)$$

where c is an absolute constant.

To prove Part 1, we observe that for all k , using the AM-GM inequality, we have that for all $Q \in \mathcal{Q}$,

$$\sum_{i < k} (\mathcal{E}_h(\pi^i, Q))^2 \leq 2 \sum_{i < k} (\widehat{\mathcal{E}}_h^i(Q))^2 + 2 \sum_{i < k} (\mathcal{E}_h(\pi^i, Q) - \widehat{\mathcal{E}}_h^i(Q))^2.$$

For $Q \in \mathcal{Q}^k$, the definition of $\mathcal{Q}^k$ implies that $\sum_{i < k} (\widehat{\mathcal{E}}_h^i(Q))^2 \leq \beta$, while (7.28) implies that $\sum_{i < k} (\mathcal{E}_h(\pi^i, Q) - \widehat{\mathcal{E}}_h^i(Q))^2 \lesssim \beta$, which gives the result.

For Part 2, we similarly observe that for all k , h and $Q \in \mathcal{Q}$,

$$\sum_{i < k} (\widehat{\mathcal{E}}_h^i(Q))^2 \leq 2 \sum_{i < k} (\mathcal{E}_h(\pi^i, Q))^2 + 2 \sum_{i < k} (\mathcal{E}_h(\pi^i, Q) - \widehat{\mathcal{E}}_h^i(Q))^2.$$

Since $Q^{M^*, \star}$ has $\mathcal{E}_h(\pi, Q^{M^*, \star}) = 0 \quad \forall \pi$ by Bellman optimality, we have

$$\sum_{i < k} (\widehat{\mathcal{E}}_h^i(Q^{M^*, \star}))^2 \leq 2 \sum_{i < k} (\mathcal{E}_h(\pi^i, Q^{M^*, \star}) - \widehat{\mathcal{E}}_h^i(Q^{M^*, \star}))^2 \leq 2C^2 \frac{\log(|\mathcal{Q}|HK/\delta)}{n},$$

where the last inequality uses (7.28). It follows that as long as $\beta \geq 2C^2 \frac{\log(|\mathcal{Q}|HK/\delta)}{n}$, we will have $Q^{M^*, \star} \in \mathcal{Q}^k$ for all k . $\square$

The next result shows that whenever the event in the previous lemma holds, the value functions constructed by BiLinUCB are optimistic.

Lemma 30: Whenever the event in Lemma 29 occurs, the following properties hold:

1. Define

$$\Sigma_h^k = \sum_{i < k} X_h^{M^*}(\pi^i) X_h^{M^*}(\pi^i)^\top. \quad (7.29)$$

For all $k \in [K]$, all $Q \in \mathcal{Q}^k$ satisfy

$$\|W_h^{M^*}(Q)\|_{\Sigma_h^k}^2 \lesssim \beta. \quad (7.30)$$

2. For all k , Q^k is optimistic in the sense that

$$\mathbb{E}_{s_1 \sim d_1}[Q_1^k(s_1, \pi_Q(s_1))] \geq \mathbb{E}_{s_1 \sim d_1}[Q_1^{M^*,*}(s_1, \pi_{M^*}(s_1))] = f^{M^*}(\pi_{M^*}). \quad (7.31)$$

Proof of Lemma 30. For Part 1, recall that by the Bilinear class property, we can write $\mathcal{E}_h(\pi^k, Q) = \langle X^{M^*}(\pi^k), W^{M^*}(Q) \rangle$, so that (7.27) implies that

$$\|W^{M^*}(Q)\|_{\Sigma_h^k} = \sum_{i < k} \langle X^{M^*}(\pi^i), W^{M^*}(Q) \rangle^2 = \sum_{i < k} (\mathcal{E}_h(\pi^i, Q))^2 \lesssim \beta.$$

For Part 2, we observe that for all k , since $Q^{M^*,*} \in \mathcal{Q}^k$, we have

$$\begin{aligned} \mathbb{E}_{s_1 \sim d_1}[Q_1^k(s_1, \pi_Q(s_1))] &= \sup_{Q \in \mathcal{Q}} \mathbb{E}_{s_1 \sim d_1}[Q_1(s_1, \pi_Q(s_1))] \\ &\geq \mathbb{E}_{s_1 \sim d_1}[Q_1^{M^*,*}(s_1, \pi_{M^*}(s_1))] \\ &= f^{M^*}(\pi_{M^*}). \end{aligned}$$

□

Proving the main result. Equipped with the lemmas above, we prove Proposition 47.

Proof of Proposition 47. We first prove a generic bound on the suboptimality of each policy π^k for $k \in [K]$. Let us condition on the event in Lemma 29, which occurs with probability at least $1 - \delta$. Whenever this event occurs, Lemma 30 implies that Q^k is optimistic, so we have can bound

$$f^{M^*}(\pi_{M^*}) - f^{M^*}(\pi^k) \leq \mathbb{E}_{s_1 \sim d_1}[Q_1^k(s_1, \pi_{Q^k}(s_1))] - f^{M^*}(\pi^k) \quad (7.32)$$

$$= \sum_{h=1}^H \mathbb{E}^{M^*, \pi^k} \left[Q_h^k(s_h, a_h) - r_h - \max_{a \in \mathcal{A}} Q_{h+1}^k(s_{h+1}, a) \right] \quad (7.33)$$

$$= \sum_{h=1}^H \langle X_h^{M^*}(\pi^k), W_h^{M^*}(Q^k) \rangle, \quad (7.34)$$

where the first equality uses the Bellman residual decomposition (Lemma 14), and the second inequality uses the Bellman rank assumption. For any $\lambda \geq 0$, using Cauchy-Schwarz, we can bound

$$\sum_{h=1}^H \langle X_h^{M^*}(\pi^k), W_h^{M^*}(Q^k) \rangle \leq \sum_{h=1}^H \|X_h^{M^*}(\pi^k)\|_{(\lambda I + \Sigma_h^k)^{-1}} \|W_h^{M^*}(Q^k)\|_{\lambda I + \Sigma_h^k}$$

For each $h \in [H]$, applying the bound in (7.30) gives

$$\|W_h^{M^*}(Q^k)\|_{\lambda I + \Sigma_h^k} \leq \sqrt{\lambda \|W_h^{M^*}(Q^k)\|_2^2 + \beta} \leq \lambda^{1/2} + \beta^{1/2},$$

where we have used that $\|W_h^{M^*}(Q^k)\|_2 \leq 1$ by assumption. This allows us to bound

$$\sum_{h=1}^H \langle X_h^{M^*}(\pi^k), W_h^{M^*}(Q^k) \rangle \lesssim (\lambda^{1/2} + \beta^{1/2}) \cdot \sum_{h=1}^H \|X_h^{M^*}(\pi^k)\|_{(\lambda I + \Sigma_h^k)^{-1}}. \quad (7.35)$$

If we can find a policy π^k for which the right-hand side of (7.35) is small, this policy will be guaranteed to have low regret. The following lemma shows that such a policy is guaranteed to exist.

Lemma 31: For any $\lambda > 0$, as long as $K \geq Hd \log(1 + \lambda^{-1}K/d)$, there exists $k \in [K]$ such that

$$\|X_h^{M^*}(\pi^k)\|_{(\lambda I + \Sigma_h^k)^{-1}}^2 \lesssim \frac{Hd \log(1 + \lambda^{-1}K/d)}{K} \quad \forall h \in [H]. \quad (7.36)$$

We choose $\lambda = \beta$, which implies that it suffices to take $K \gtrsim Hd \log(1 + n/d)$ to satisfy the condition in Lemma 31. By choosing k to satisfy (7.36) and plugging this bound into (7.35), we conclude that the policy π^k has

$$f^{M^*}(\pi_{M^*}) - f^{M^*}(\pi^k) \lesssim H \sqrt{\beta \cdot \frac{Hd \log(1 + \beta^{-1}K/d)}{K}} \lesssim \tilde{O}\left(H^{3/2} \sqrt{\frac{d \log(|\mathcal{Q}|/\delta)}{n}}\right) \lesssim \varepsilon \quad (7.37)$$

as desired.

Finally, we need to argue that the policy $\hat{\pi}$ returned by the algorithm is at least as good as π^k . This is straightforward and we only sketch the argument: By Hoeffding's inequality and a union bound, we have that with probability at least $1 - \delta$, for all k ,

$$|f^{M^*}(\pi^k) - \hat{V}^k| \lesssim \sqrt{\frac{\log(K/\delta)}{n}},$$

which implies that $f^{M^*}(\hat{\pi}) \gtrsim f^{M^*}(\pi^k) - \sqrt{\frac{\log(K/\delta)}{n}}$. The error term here is of lower order than (7.37). $\square$

Deferred proofs. To finish up, we prove Lemma 31.

Proof of Lemma 31. To prove the result, we need a variant of the elliptic potential lemma (Lemma 11).

Lemma 32 (e.g. Lemma 19.4 in Lattimore and Szepesvári [60]): Let $a_1, \dots, a_T \in \mathbb{R}^d$ satisfy $\|a_t\|_2 \leq 1$ for all $t \in [T]$. Fix $\lambda > 0$, and let $V_t = \lambda I + \sum_{s < t} a_s a_s^\top$. Then

$$\sum_{t=1}^T \log(1 + \|a_t\|_{V_t^{-1}}^2) \leq d \log(1 + \lambda^{-1}T/d). \quad (7.38)$$

For any $\lambda > 0$, applying this result for each $h \in [H]$ and summing gives

$$\sum_{k=1}^K \sum_{h=1}^H \log(1 + \|X_h^{M^*}(\pi^k)\|_{(\lambda I + \Sigma_h^k)^{-1}}^2) \leq Hd \log(1 + \lambda^{-1}K/d).$$

This implies that there exists k such that

$$\sum_{h=1}^H \log(1 + \|X_h^{M^*}(\pi^k)\|_{(\lambda I + \Sigma_h^k)^{-1}}^2) \leq \frac{Hd \log(1 + \lambda^{-1}K/d)}{K},$$

which means that for all $h \in [H]$, $\log(1 + \|X_h^{M^*}(\pi^k)\|_{(\lambda I + \Sigma_h^k)^{-1}}^2) \leq \frac{Hd \log(1 + \lambda^{-1}K/d)}{K}$, or equivalently:

$$\|X_h^{M^*}(\pi^k)\|_{(\lambda I + \Sigma_h^k)^{-1}}^2 \leq \exp\left(\frac{Hd \log(1 + \lambda^{-1}K/d)}{K}\right) - 1.$$

As long as $K \geq Hd \log(1 + \lambda^{-1}K/d)$, using that $e^x \leq 1 + 2x$ for $0 \leq x \leq 1$, we have

$$\|X_h^{M^*}(\pi^k)\|_{(\lambda I + \Sigma_h^k)^{-1}}^2 \leq 2 \frac{Hd \log(1 + \lambda^{-1}K/d)}{K}.$$

□

7.3.3 Bellman Rank: Examples

We now highlight concrete examples of models with low Bellman rank. We start with familiar examples, then introduce new models that allow for nonlinear function approximation.

Example 7.1 (Tabular MDPs). If M is a tabular MDP with $|\mathcal{S}| \leq S$ and $|\mathcal{A}| \leq A$, we can write the Bellman residual for any function Q and policy π as

$$\begin{aligned} \mathcal{E}_h(\pi, Q) &= \mathbb{E}^{M, \pi} \left[Q_h(s_h, a_h) - \left(r_h + \max_a Q_{h+1}(s_{h+1}, a) \right) \right] \\ &= \sum_{s, a} d_h^{M, \pi}(s, a) \mathbb{E}^M \left[Q_h(s, a) - \left(r_h + \max_{a'} Q_{h+1}(s_{h+1}, a') \right) \mid s_h = s, a_h = a \right]. \end{aligned}$$

It follows that if we define

$$X_h^M(\pi) = \{d_h^{M, \pi}(s, a)\}_{s \in \mathcal{S}, a \in \mathcal{A}} \in \mathbb{R}^{SA}$$

and

$$W_h^M(Q) = \left\{ \mathbb{E}^M \left[Q_h(s, a) - \left(r_h + \max_{a'} Q_{h+1}(s_{h+1}, a') \right) \mid s_h = s, a_h = a \right] \right\}_{s \in \mathcal{S}, a \in \mathcal{A}} \in \mathbb{R}^{SA},$$

we have

$$\mathcal{E}_h(\pi, Q) = \langle X_h^M(\pi), W_h^M(Q) \rangle.$$

This shows that $d_B(M) \leq SA$.

◁

Example 7.2 (Low-Rank MDPs). The calculation in (7.21) shows that by choosing $X_h^M(\pi) := \mathbb{E}^{M, \pi}[\phi(s_h, a_h)] \in \mathbb{R}^d$ and $W_h^M(Q) := \theta_h^Q - w_h^M - \tilde{\theta}_h^{M, Q} \in \mathbb{R}^d$, any Low-Rank MDP M has $d_B(M) \leq d$. When specialized to this setting, the regret of BiLinUCB is worse than that of LSVI-UCB (though still polynomial in all of the problem parameters). This is because BiLinUCB is a more general algorithm, and does not take advantage of an additional feature of the Low-Rank MDP model known as *Bellman completeness*: If M is a Low-Rank MDP, then for all $Q \in \mathcal{Q}$, we have $\mathcal{T}_h^M Q_{h+1} \in \mathcal{Q}_{h+1}$. By using a more specialized relative of BiLinUCB that incorporates a modified confidence set construction to exploit completeness, it is possible to match and actually improve upon the regret of LSVI-UCB [49].

◁

We now explore Bellman rank for some MDP families that have not already been covered.

Example 7.3 (Low Occupancy Complexity). An MDP M is said to have *low occupancy complexity* if there exists a feature map $\phi^M(s, a) \in \mathbb{R}^d$ such that for all π , there exists $\theta_h^{M, \pi} \in \mathbb{R}^d$ such that

$$d_h^{M, \pi}(s, a) = \langle \phi^M(s, a), \theta_h^{M, \pi} \rangle. \quad (7.39)$$

Note that neither ϕ^M nor $\theta_h^{M, \pi}$ is assumed to be known to the learner. If M has *low occupancy complexity*, then for any value function Q and policy π , we have

$$\begin{aligned} \mathcal{E}_h(\pi, Q) &= \mathbb{E}^{M, \pi} \left[Q_h(s_h, a_h) - \left(r_h + \max_a Q_{h+1}(s_{h+1}, a) \right) \right] \\ &= \sum_{s, a} d_h^{M, \pi}(s, a) \mathbb{E}^M \left[Q_h(s, a) - \left(r_h + \max_{a'} Q_{h+1}(s_{h+1}, a') \right) \mid s_h = s, a_h = a \right] \\ &= \sum_{s, a} \langle \phi^M(s, a), \theta_h^{M, \pi} \rangle \mathbb{E}^M \left[Q_h(s, a) - \left(r_h + \max_{a'} Q_{h+1}(s_{h+1}, a') \right) \mid s_h = s, a_h = a \right] \\ &= \left\langle \theta_h^{M, \pi}, \sum_{s, a} \phi^M(s, a) \mathbb{E}^M \left[Q_h(s, a) - \left(r_h + \max_{a'} Q_{h+1}(s_{h+1}, a') \right) \mid s_h = s, a_h = a \right] \right\rangle. \end{aligned}$$

It follows that if we define

$$X_h^M(\pi) = \theta_h^{M, \pi}$$

and

$$W_h^M(Q) = \sum_{s, a} \phi^M(s, a) \mathbb{E}^M \left[Q_h(s, a) - \left(r_h + \max_{a'} Q_{h+1}(s_{h+1}, a') \right) \mid s_h = s, a_h = a \right],$$

then $\mathcal{E}_h(\pi, Q) = \langle X_h^M(\pi), W_h^M(Q) \rangle$, which shows that $d_B(M) \leq d$.

This setting subsumes tabular MDPs and low-rank MDPs, but is substantially more general. Notably, low occupancy complexity allows for *nonlinear function approximation*: As long as the occupancies satisfy (7.39), the Bellman rank is at most d for *any class $\mathcal{Q}$* , which might consist of neural networks or other nonlinear models. $\triangleleft$

We close with two more new examples.

Example 7.4 (LQR). A classical problem in continuous control is the Linear Quadratic Regulator, or LQR. Here, we have $\mathcal{S} = \mathcal{A} = \mathbb{R}^d$, and states evolve via

$$s_{h+1} = A^M s_h + B^M a_h + \zeta_h,$$

where $\zeta_h \sim \mathcal{N}(0, I)$, and $s_1 \sim \mathcal{N}(0, I)$. We assume that rewards have the form¹⁷

$$r_h = -s_h^\top Q^M s_h - a_h^\top R^M a_h$$

for matrices $Q^M, R^M \succeq 0$. A classical result, dating back to Kalman, is that the optimal controller for this system is a linear mapping of the form

$$\pi_{M, h}(s) = K_h^M s,$$

and that the value function $Q^{M, \star}(s, a) = (s, a)^\top P_h^M(s, a)$ is quadratic. Hence, it suffices to take $\mathcal{Q}$ to be the set of all quadratic functions in (s, a) . With this choice, it can be shown that $d_B(M) \leq d^2 + 1$. The basic idea is to choose $X_h^M(\pi) = (\text{vec}(\mathbb{E}^{M, \pi}[s_h s_h^\top]), 1)$ and use the quadratic structure of the value functions. $\triangleleft$

¹⁷LQR is typically stated in terms of losses; we negate because we consider rewards.

Example 7.5 (Linear Q^*/V^*). In Proposition 45, we showed that for RL with linear function approximation, assuming only that $Q^{M,*}$ is linear is not enough to achieve low regret. It turns out that if we also assume in addition that $V^{M,*}$ is linear, the situation improves.

Consider an MDP M . Assume that there known feature maps $\phi(s, a) \in \mathbb{R}^d$ and $\psi(s') \in \mathbb{R}^d$ such that

$$Q_h^{M,*}(s, a) = \langle \phi(s, a), \theta_h^M \rangle, \quad \text{and} \quad V_h^{M,*}(s) = \langle \psi(s), w_h^M \rangle.$$

Let

$$\mathcal{Q} = \left\{ Q \mid Q_h(s, a) = \langle \phi(s, a), \theta_h \rangle : \theta_h \in \mathbb{R}^d, \exists w \max_{a \in \mathcal{A}} \langle \phi(s, a), \theta_h \rangle = \langle \psi(s), w \rangle \forall s \right\}.$$

Then $d_B(M) \leq 2d$. We will not prove this result, but the basic idea is to choose $X_h^M(\pi) = \mathbb{E}^{M, \pi}[(\phi(s_h, a_h), \psi(s_{h+1}))]$. $\triangleleft$

See Jiang et al. [46], Du et al. [34] for further examples.

7.3.4 Generalizations of Bellman Rank

While we gave (7.24) as the definition for Bellman rank, there are many variations on the assumption that also lead to low regret. One well-known variant is *V-type Bellman rank* [46], which asserts that for all $\pi \in \Pi$ and $Q \in \mathcal{Q}$, *또는 정책과 가치함수에 대해*,

$$\mathbb{E}^{M, \pi} \mathbb{E}_{s_{h+1} | s_h, a_h \sim \pi_Q(s_h)}^M \left[Q_h(s_h, a_h) - r_h - \max_{a \in \mathcal{A}} Q_{h+1}(s_{h+1}, a) \right] = \langle X_h^M(\pi), W_h^M(Q) \rangle. \quad (7.40)$$

This is the same as the definition (7.24) (which is typically referred to as *Q-type Bellman rank*), except that we take $a_h = \pi_Q(s_h)$ instead of $a_h = \pi(s_h)$.¹⁸ With an appropriate modification, BiLinUCB can be shown to give sample complexity guarantees that scale with *V-type Bellman rank* instead of *Q-type*. This definition captures meaningful classes of tractable RL models [that are not captured by the *Q-type* definition (7.24)], with a canonical example being *Block MDPs*.

Example 7.6 (Block MDP). The Block MDP [46, 33, 64] is a model in which the (“observed”) state space $\mathcal{S}$ is large/high-dimensional, but the dynamics are governed by a (small) latent state space $\mathcal{Z}$. Formally, a Block MDP $M = (\mathcal{S}, \mathcal{A}, P, R, H, d_1)$ is defined based on an (unobserved) latent state space $\mathcal{Z}$, with z_h denoting the latent state at layer h . We first describe the dynamics for the latent space. Given initial latent state z_1 , the latent states evolve via

$$z_{h+1} \sim P_h^{M, \text{latent}}(z_h, a_h).$$

prob. dist.

The latent state z_h is not observed. Instead, we observe

$$s_h \sim q_h^M(z_h),$$

emission dist.

where $q_h^M : \mathcal{Z} \rightarrow \Delta(\mathcal{S})$ is an *emission distribution* with the property that $\text{supp}(q_h(z)) \cap \text{supp}(q_h(z')) = \emptyset$ if $z \neq z'$. This property (*decodability*) ensures that there exists a *unique*

¹⁸The name “V-type” refers to the fact that (7.40) only depends on Q through the induced V -function $s \mapsto Q_h(s, \pi_Q(s))$, while (7.24) depends on the full Q -function, hence “Q-type”.

unique mapping $\phi_h^M : \mathcal{S} \rightarrow \mathcal{Z}$ that maps the observed state s_h to the corresponding latent state z_h . We assume that $R_h^M(s, a) = R_h^M(\phi_h^M(s), a)$, which implies that optimal policy π_M depends only on the endogenous latent state, i.e. $\pi_{M,h}(s) = \pi_{M,h}(\phi_h^M(s))$.

The main challenge of learning in Block MDPs is that the decoder ϕ^M is not known to the learner in advance. Indeed, given access to the decoder, one can obtain regret $\text{poly}(H, |\mathcal{Z}|, |\mathcal{A}|) \cdot \sqrt{T}$ by applying tabular reinforcement learning algorithms to the latent state space. In light of this, the aim of the Block MDP setting is to obtain sample complexity guarantees that are independent of the size of the observed state space $|\mathcal{S}|$, and scale as $\text{poly}(|\mathcal{Z}|, |\mathcal{A}|, H, \log|\mathcal{F}|)$, where $\mathcal{F}$ is an appropriate class of function approximators (typically either a value function class $\mathcal{Q}$ containing $Q^{M,*}$ or a class of decoders Φ that attempts to model ϕ^M directly).

We now show that the Block MDP setting admits low V-type Bellman rank. Observe that we can write

$$\begin{aligned} & \mathbb{E}^{M, \pi} \mathbb{E}_{s_{h+1}|a_h, s_h \sim \pi_Q(s_h)}^M \left[Q_h(s_h, \pi_Q(s_h)) - r_h - \max_{a \in \mathcal{A}} Q_{h+1}(s_{h+1}, \pi_Q(s_{h+1})) \right] \\ &= \sum_{z \in \mathcal{Z}} d_h^{M, \pi}(z) \mathbb{E}_{s \sim q_h^M(z)}^M \mathbb{E}_{s_{h+1}|s_h, a_h \sim \pi_Q(s_h)}^M \left[Q_h(s_h, \pi_Q(s_h)) - r_h - \max_{a \in \mathcal{A}} Q_{h+1}(s_{h+1}, \pi_Q(s_{h+1})) \right]. \end{aligned}$$

This implies that we can take

$$X_h^M(\pi) = \{d_h^{M, \pi}(z)\}_{z \in \mathcal{Z}}$$

and

$$W_h^M(Q) = \left\{ \mathbb{E}_{s \sim q_h^M(z)}^M \mathbb{E}_{s_{h+1}|s_h, a_h \sim \pi_Q(s_h)}^M \left[Q_h(s_h, \pi_Q(s_h)) - r_h - \max_{a \in \mathcal{A}} Q_{h+1}(s_{h+1}, \pi_Q(s_{h+1})) \right] \right\}_{z \in \mathcal{Z}}$$

so that the V-type Bellman rank is at most $|\mathcal{Z}|$. This means that as long as $\mathcal{Q}$ contains $Q^{M,*}$, we can obtain sample complexity guarantees that scale with $|\mathcal{Z}|$ (rather than $|\mathcal{S}|$) as desired. $\triangleleft$

(There are a number of variants of Bellman rank, including Bilinear rank [34] and Bellman-Eluder dimension [49], which subsume and slightly generalize both Bellman rank definitions.)

7.3.5 Decision-Estimation Coefficient for Bellman Rank

An alternative to the BiLinUCB method is to appeal to the E2D meta-algorithm and the Decision-Estimation Coefficient. The following result [40] shows that the Decision-Estimation Coefficient is always bounded for classes with low Bellman rank.

Proposition 48: For any class of MDPs $\mathcal{M}$ (for which all $M \in \mathcal{M}$ have Bellman rank at most d) we have

$$\text{dec}_\gamma(\mathcal{M}) \lesssim \frac{H^2 d}{\gamma}. \quad (7.41)$$

low Bellman rank d
decision estim. coeff. is always bounded

This implies that that E2D meta-algorithm has $\mathbb{E}[\mathbf{Reg}] \lesssim H\sqrt{dT \cdot \mathbf{Est}_H}$ whenever we have access to a realizable model class (with low Bellman rank). As a special case, for any finite class $\mathcal{M}$, using averaged exponential weights as an estimation oracle gives

$$\mathbb{E}[\mathbf{Reg}] \lesssim H\sqrt{dT \log|\mathcal{M}|}. \quad (7.42)$$

We will not prove Proposition 48, but interested readers can refer to Foster et al. [40]. The result can be proven using two approaches, both of which build on the techniques we have already covered. The first approach is to apply a more general version of the PC-IGW algorithm from Section 6.6, which incorporates optimal design in the space of policies. The second approach is to move to the Bayesian DEC and appeal to posterior sampling, as in Section 4.4.2.

Value-based guarantees via optimistic estimation. In general, the model estimation complexity $\log|\mathcal{M}|$ in (7.42) can be arbitrarily large compared to the complexity $\log|\mathcal{Q}|$ for a realizable value function class (consider the low-rank MDP—since μ is unknown, it is not possible to construct a small model class $\mathcal{M}$). To derive value-based guarantees along the lines of what BiLinUCB achieves in Proposition 47, a natural approach is to replace the Hellinger distance (appearing in the DEC) with a divergence tailored to value function iteration, following the development in Sections 6.7.2 and 6.7.3. Once such choice is the divergence

Hellinger distance RMS
Value Iteration on tailored
divergence is 48
** Asymmetric.*

$$D_{\text{sbr}}^\pi(Q \| M) = \sum_{h=1}^H \left(\mathbb{E}^{M, \pi} \left[Q_h(s_h, a_h) - \left(r_h + \max_a Q_{h+1}(s_{h+1}, a) \right) \right] \right)^2,$$

which measures the squared bellman residual for an estimated value function under M . With this choice, we appeal to the optimistic E2D algorithm (E2D.Opt) from Section 6.7.3. One can show that the optimistic DEC for this divergence is bounded as

$$\text{o-dec}_\gamma^{D_{\text{sbr}}^\pi}(\mathcal{M}) \lesssim \frac{H \cdot d}{\gamma}.$$

This implies that E2D.Opt, with an appropriate choice of estimation algorithm tailored to $D_{\text{sbr}}^\pi(\cdot \| \cdot)$, achieves

$$\mathbb{E}[\mathbf{Reg}] \lesssim (H^2 d \log|\mathcal{Q}|)^{1/2} T^{3/4}.$$

Note that due to the asymmetric nature (of $D_{\text{sbr}}^\pi(\cdot \| \cdot)$), it is critical to appeal to optimistic estimation to derive this result. Indeed, the non-optimistic generalized DEC $\text{dec}_\gamma^{D_{\text{sbr}}^\pi}$ does not enjoy a polynomial bound. See Foster et al. [41] for details.

[Note: This subsection will be expanded in the next version.]